

DONGHYUN SOHN

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EDUCATION

Northwestern University

Evanston, IL, United States

Ph.D. in Computer Science

Sep. 2022 - Present

- Advisor: Jennie Rogers

Sogang University

Seoul, Republic of Korea

B.S. in Computer Science and Engineering

Mar. 2016 - Aug. 2021

- Summa Cum Laude (2 years of military service included)

EXPERIENCE

Graduate Research Assistant, Northwestern University

Evanston, IL, United States

Sep. 2022 - Present

- Introduced Alchemy, a protocol-agnostic optimization framework for privacy-preserving query processing, combining traditional optimization techniques with circuit-aware cost modeling. Achieved up to $100\times$ speedups in secure query execution across multiple cryptographic protocols.
- Engineered an OpenFHE-based columnar engine; optimized secure query latency via SW/HW co-design (SIMD, multi-threading) and hybrid FHE-to-MPC conversion.
- Architected a hierarchical graph-based alignment for NL2SQL; optimized complex schema mapping and join-path discovery through structural graph modeling.
- Designed a cost-aware optimizer for secure SQL workloads with ML-based runtime prediction, incorporating AWS hardware characteristics for performance- and cost-efficient instance selection.
- Built a Differential Privacy-compliant data synthesis framework for OLAP performance regression, ensuring high fidelity simulation of statistical distributions and physical storage layouts in DuckDB/Parquet.

Undergraduate Research Intern, Sogang University

Seoul, Republic of Korea

Aug. 2020 - Dec. 2021

- Optimized the BIRCH clustering algorithm by designing a disk-based Clustering Feature tree with selective buffering, reducing disk I/O and improving clustering quality by 7% for large-scale datasets.

PUBLICATIONS

1. **Donghyun Sohn**, Kelly Jiang, Nicolas Hammer, Jennie Rogers, “Alchemy: A Query Optimization Framework for Oblivious SQL,” *Proceedings of the VLDB Endowment*, 2025. [Paper] [Code]
2. **Donghyun Sohn**, Xiling Li, Jennie Rogers, “Everything You Always Wanted to Know About Secure and Private Database Systems (but were Afraid to Ask),” *Data Engineering Bulletin*, 2023. [Paper]
3. **Donghyun Sohn**, Sungwon Jung, “Disk-based BACF Tree for Clustering Massive Datasets,” *Korea Software Congress*, 2021. [Paper] [Code]

SERVICES

Program Committee Member

SIGMOD Availability & Reproducibility Initiative (ARI)

2024, 2025

Teaching Assistant, Northwestern University

COMP_SCI 339: Intro to Database Systems

Fall 2023, 2024, 2025

SKILLS

Programming: C, C++, Java, Python, SQL

Systems/Tools: AWS, Docker, Ubuntu, EMP-Toolkit, OpenFHE